Local goodness-of-fit measures for neural decoding

Sirui Zeng[1], Alison E. Comrie[5], Loren M. Frank[2,3,4],
Uri T. Eden[1], Eric L. Denovellis[2,4,*]

[1]Department of Mathematics and Statistics, Boston University, Boston, MA, USA

[2]Departments of Physiology and Psychiatry, UCSF, San Francisco, CA 94158, USA

[3]Kavli Institute for Fundamental Neuroscience, UCSF, San Francisco, CA 94158, USA

[4]HHMI, UCSF, San Francisco, CA 94158, USA

[5]HHMI Janelia Research Campus, Ashburn, VA 20147, USA

[*]Corresponding author. Email: eric.denovellis@ucsf.edu

Abstract

State-space models are widely used in many fields to infer latent states from observations, but there are still some issues and challenges related to them. First, because latent variables are inherently unobservable, their decoding results cannot be directly validated against observed data. Second, it is often challenging to disentangle prediction errors stemming from model misspecification. Third, such errors can accumulate and propagate over time, amplifying uncertainty in long-term predictions. Moreover, many state-space models are purposefully simplified, accepting some degree of inaccuracy in exchange for greater computational efficiency. To address this challenge, we developed three classes of metrics designed to identify localized periods where model misspecification leads to inaccurate estimates. Each metric assesses the consistency between the information contained in individual observations and the information extracted by the state-space model from a local window of data. These metrics fall into three categories: 1) a set of divergence measures, specifically

F-divergences, which compare the distributions of the latent state process directly; 2) measures of overlap between estimated credible regions obtained from the state distributions; and 3) predictive checks for individual observations, based on sample estimated from the assumed state-space model. We applied these metrics to the problem of estimating nonlocal spatial representations in both simulated and real hippocampal spiking data, demonstrating the strengths and limitations of our approach.

Keywords: Neural decoding, State-space models, Goodness-of-fit

1 Introduction

Many brain states, such as those related to memory, emotion, and attention, are not directly observable. They are internal states that are not solely determined by the external world and can change over time. This makes them challenging to understand. However, these brain states are critical for understanding how the brain supports complex behavior, because they go beyond simple stimulus and response relationships. A major challenge for systems neuroscience is to relate neural activity (spikes, spike waveforms, calcium traces, etc.) to these latent dynamic brain states.

State space models are a powerful way to understand how neural activity relates to latent dynamic brain states. They have been applied to diverse neural systems such as gustatory activity [Jones et al., 2007], hippocampal replay dynamics [Denovellis et al., 2021], and hypothalamic activity dynamics [Vinograd et al., 2024]. The core assumption of these state space models is that complex, high dimensional neural activity can be related to low dimensional, latent states. State space models aggregate information over time about the latent states, weighing both the information from the data at the current time (normalized likelihood distribution) with the accumulated information used to predict the latent state (prediction distribution). This powerful combination enables robust estimation of the latent state, effectively balancing current information with the accumulated history of the system and prior knowledge.

Despite the wide use of state space models in neuroscience, methods to validate state space models remain underdeveloped and underutilized. One reason is that latent states are inherently difficult to evaluate [Newman et al., 2014]. Without ground truth, model validity can only be inferred indirectly—from relative likelihoods under multiple models, from the quality of reconstructed signals, or from performance on downstream tasks. Another challenge is that neural data are dynamic, so model fit may vary substantially across time. Compounding this problem is that common methods of validating models, such as cross-validation and "shuffle" permutation tests, are holistic evaluations of the model – they tell neuroscientists whether or not the model generalizes well or is different from a simplified model with less utility – but they fail to indicate when the model fits poorly or which specific components are responsible. Consequently, these methods can reject models that are intentionally simplified yet scientifically meaningful and offer little guidance for targeted improvement of the model. Many state-space models in neuroscience are intentionally simplified—not because the brain is simple, but because tractable computation and clear interpretation often matter most. For example, some state-space models smooth emotional movement into a low-dimensional trajectory [Deng et al., 2015]; others compress field-linked structure into a stationary, locally anchored map [Hsin et al., 2022]. Some frameworks simplify both at once, jointly compressing affective trajectories and local field potentials into compact latent states for computation and inference [Tao et al., 2018, Zeng and Eden, 2025]. Under standard goodness-of-fit criteria, such simplifications would likely fail.

State space models already contain rich diagnostic information in the form of distributions used in their estimation. One potential solution for more targeted local diagnostic information of state space models would be to utilize the information in the normalized likelihood, which reflects information from only the current neural observations, and prediction distribution, which aggregates information from all past observations based on the state space model structure. When these two distributions agree, the model's prior expectations and data-driven evidence are consistent. When they diverge, the mismatch can reveal where and when and potentially how the model fails to capture the structure of the data. Examining these discrepancies could therefore provide a direct, interpretable measure of model–data agreement—analogous to a local, time-resolved goodness-of-fit test.

To address these gaps, we develop and validate a suite of three complementary goodness-of-fit methods – KL divergence, highest posterior density (HPD) region overlap, predictive checks – designed to provide actionable feedback to neuroscientists. Specifically, each of these methods utilizes the mismatch between prediction and likelihood to: (1) identify time periods of poor model–data agreement; (2) distinguish errors arising from the state versus observation components; and (3) provide intuitive visualizations accessible to a broad audience. We demonstrate the utility of these techniques on simulated and real hippocampal data, offering a path for neuroscientists to move from binary model rejection toward an iterative, diagnostic approach to building interpretable theories of neural dynamics.

2 Methods

2.1 State-Space Models

The state-space paradigm is particularly well suited for neural data analysis since many neuroscience experiments focus on internal brain processes that are not directly observable. For example, hippocampal replay of spatial trajectories has been hypothesized to be related to mental exploration of space, but the specific spatial trajectories being replayed are not directly observable by experimentalists [Tao et al., 2018]. In such cases, state-space models are used to estimate simultaneously the unobserved internal representations and their relation to neural activity patterns.

For a discrete set of time points, $t_0, \ldots, t_K$, we define an unobserved state process x_k, which influences a neural observation process y_k at each time t_k. We define a state space model by combining a state transition distribution $p(x_k \mid x_{k-1})$, which defines how the state evolves in time, with an observation likelihood $p(y_k \mid x_k)$, which defines the neural representation of each state. To simplify the problem, we assume that the state transition depends only on the most recent past value of the state and that the observation likelihood depends only on the current state value. Figure 1(a) shows a schematic of this state-space model structure.

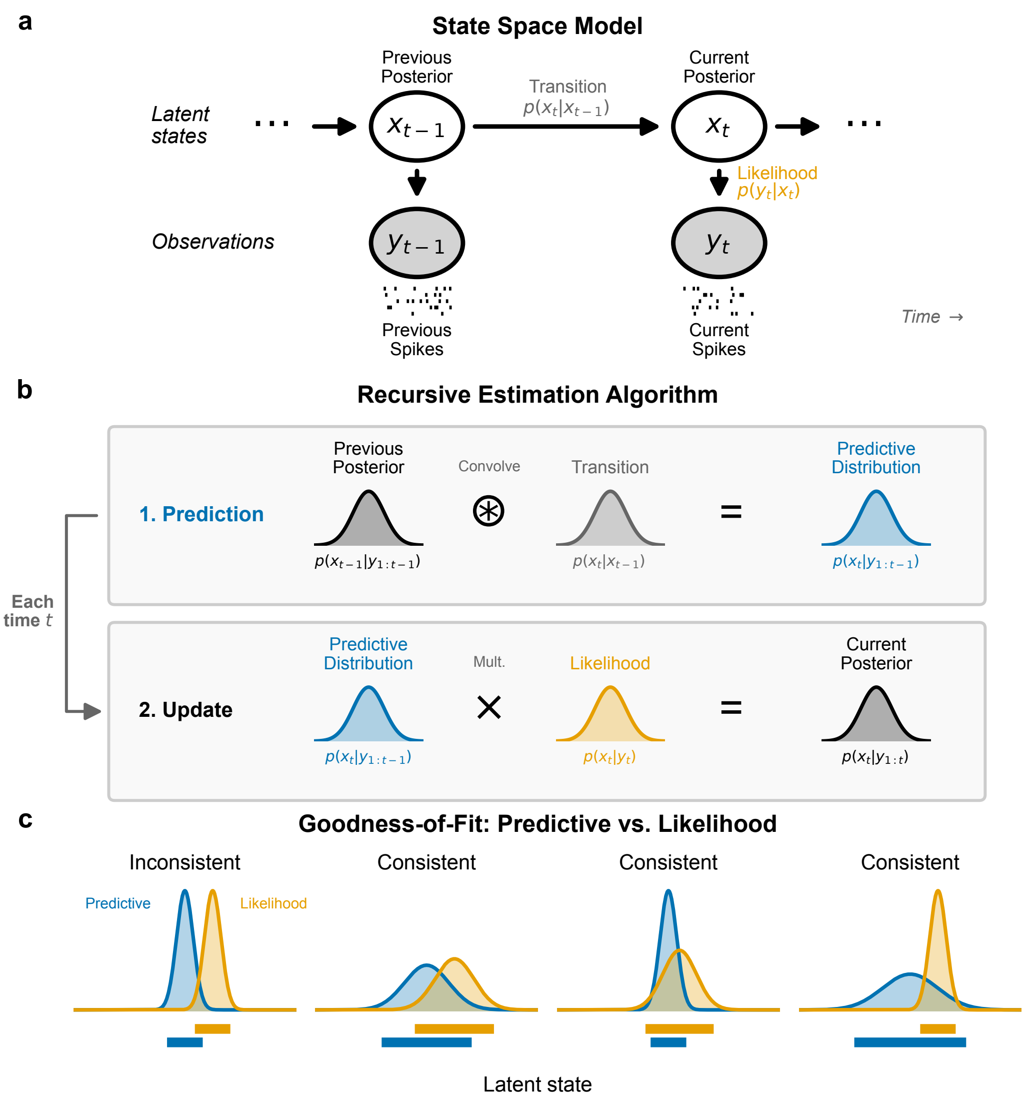

Figure 1: Schematics for state-space model and goodness-of-fit measures. (a) Graphical model depiction of state-space model includes state transition model, $p(x_k \mid x_{k-1})$, and observation likelihood $p(y_k \mid x_k)$. (b) These model components are sufficient to compute an iterative expression for the posterior filter distribution at time t_k, $p(x_k \mid y_{1:k})$ from the distribution at the previous time step. This involves first convolving the previous posterior with the state transition distribution to compute the one-step predictive distribution, $p(x_k \mid y_{1:k-1})$, and then updating the predictive distribution with information from the most recent spiking by multiplying by the observation likelihood. (c) For each new spike observation, we compute the local goodness-of-fit by assessing the consistency of information between the predictive distribution and likelihood. These are considered inconsistent when they assign high probability to mostly disjoint subsets of the state-space. When the regions with high probability overlap, the distributions are considered consistent, even if the distributions differ in their central tendency or certainty.

This state-space model structure allows for iterative estimation of the state process at each time step based on all the neural data up to that time, as shown in Figure 1(b). If the distribution of the state at time t_{k-1} has been computed, we can convolve it with the state transition distribution to obtain a one-step prediction distribution, $p(x_k \mid y_{1:k-1}) = \int_{x_{k-1}} p(x_k \mid x_{k-1}) \cdot p(x_{k-1} \mid y_{1:k-1})\, dx_{k-1}$. This predictive distribution combines the information from all previous observations to compute the distribution of the state at the next time step. We update this predictive distribution with the information from the most recent observation by multiplying by the observation likelihood to compute the posterior distribution of the state given all of the data up to the current time step, $p(x_k \mid y_{1:k}) \propto p(y_k \mid x_k) \cdot p(x_k \mid y_{1:k-1})$.

Common approaches for assessing the goodness-of-fit of state-space models include examining the cross-validated error measures in the state process when the state is observable [Stone, 1974], cross-validated error measures in the observation process such as the mean-squared prediction error [Hastie et al., 2009], and the distribution of prediction residuals [Shen and Zhuang, 2024]. For many neural modeling problems, state-space models are intentionally simplified and these standard goodness-of-fit measures trivially show lack of fit.

Here we explore a set of local goodness-of-fit measures, each of which compares the prediction distribution to the normalized likelihood of the most recent observation as a function of the state or to that observation itself. Thus, these measures consider the consistency between information accumulated through the state-space model over past spiking with the information from the most recent spiking activity. We do not expect these distributions to align perfectly since they likely contain different amounts of information about the state process. Instead, we assess whether the information these distributions provide about the state and observation processes is consistent in the sense illustrated in Figure 1(c). Specifically, we say that the two distributions are inconsistent when their probability mass is concentrated in mostly non-overlapping regions of the state space. This suggests disagreement between the information from past spiking and the information from the most recent spiking. Distributions that largely align, or that are shifted from each other but are supported over largely overlapping regions of state space are considered consistent. Additionally, if one distribution covers a broad region of the state-space and the other covers a much narrower subset of that region, the distributions are considered to be consistent. We expect this situation to arise often since the prediction distribution contains information accumulated from many past spikes.

2.2 Local Goodness-Of-Fit Measures

In this section, we introduce three classes of metrics designed to characterize the local goodness-of-fit of the state-space model.

2.2.1 KL Divergence

One of the most well-known measures of discrepancy between probability distributions is the Kullback–Leibler (KL) divergence [Kullback and Leibler, 1951], defined in this case as the expected excess self-information (log of the density function) incurred when the likelihood is used to approximate the predictive distribution, with the expectation taken with respect to the predictive distribution,

$$D_{\mathrm{KL}}(P \parallel Q) = \int P(x_k) \log\left(\frac{P(x_k)}{Q(x_k)}\right) dx_k, \tag{1}$$

where $P(x_k) = p(x_k \mid y_{1:k-1})$ is the predictive distribution and $Q(x_k) = p(y_k \mid x_k)/\int p(y_k \mid x_k)\, dx_k$ is the normalized likelihood in x_k given the observed spiking y_k.

Figure 2(a) illustrates the computation of the KL divergence. The top panel shows the predictive distribution and normalized observation likelihood as functions of x_k. The middle panel shows the difference in self-information between these distributions, $\log(P(x_k)/Q(x_k))$. The bottom panel is this difference in self-information weighted by $P(x_k)$, which when integrated gives the value of the KL divergence.

The KL divergence is an asymmetric measure, meaning that $D_{\mathrm{KL}}(P \parallel Q) \neq D_{\mathrm{KL}}(Q \parallel P)$. We could also compute the KL divergence from the likelihood to the predictive distribution $D_{\mathrm{KL}}(Q \parallel P)$ or the average of these two divergences, $\frac{1}{2}(D_{\mathrm{KL}}(P \parallel Q) + D_{\mathrm{KL}}(Q \parallel P))$. More broadly, the KL divergence is part of a larger family of divergence metrics between distributions, known as F-

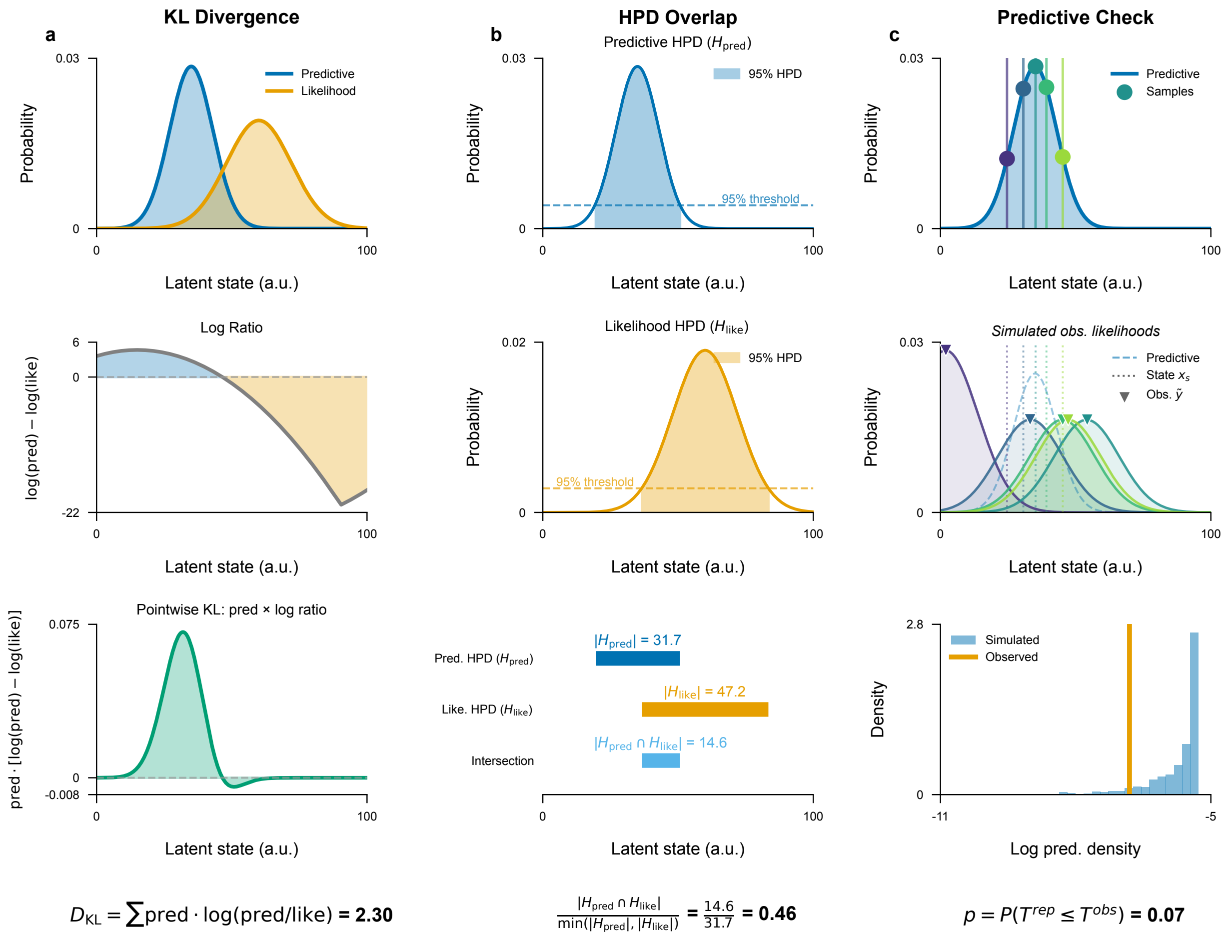

Figure 2: Visualizations of the computation of each local goodness-of-fit metric. (a) The KL divergence compares the predictive distribution (blue) and the observation likelihood (orange) (top panel) by computing the difference in their log density (middle panel) and computing its expectation with respect to the predictive distribution (bottom panel). (b) The HPD overlap is defined by computing the HPD for the predictive distribution (top) and for the observation likelihood (middle) and then computing the size of their intersection relative to the size of the smaller of these two HPD regions (bottom). (c) Predictive checks are computed by sampling values of the state from the predictive distribution multiple times (top), then sampling from the conditional distribution of neurons that could have generated the observed spike given each sampled state value (middle), and then computing the probability that the log of the predictive distribution of the observed spike is greater than the log of empirical predictive distribution of the sampled spikes (bottom).

divergences. Other members include the total variation distance [Bhattacharyya et al., 2023] and the Pearson χ^2 divergence [Crack, 2018]. It is simple to extend the KL divergence metric between the predictive distribution and the observation likelihood to any of these other divergence metrics to identify differences in additional features of the distributions.

2.2.2 Overlap Between HPD Regions

Another way to assess the consistency between the predictive distribution and the observation likelihood is to examine the extent to which their highest posterior density (HPD) regions overlap. A substantial intersection suggests that the two distributions convey consistent information, reinforcing the reliability of the inferred latent state. In contrast, minimal or no overlap signals a significant discrepancy, hinting at potential inconsistencies in the local fit of the model. This approach provides an interpretable and intuitive metric of the degree of consistency between these distributions.

We define the overlap between the HPD for the predictive distribution and the likelihood as

$$\mathrm{HPDOverlap}(P, Q) = \frac{\left|\mathrm{HPD}_P \cap \mathrm{HPD}_Q\right|}{\min\left\{\left|\mathrm{HPD}_P\right|, \left|\mathrm{HPD}_Q\right|\right\}}, \tag{2}$$

where HPD_P and HPD_Q are the highest posterior density regions of the predictive density and the normalized observation likelihood respectively, and $|\cdot|$ denotes the volume of the region. The numerator is the size of the overlap between these regions and the denominator is the size of the smaller of the two HPD regions. This normalization ensures that the resulting value is in the range $[0, 1]$, with 0 indicating no overlap, and 1 indicating that one of the HPDs is entirely contained within the other. This means that if, for example, the likelihood produces an HPD that is wider and therefore less informative than the predictive density but that completely contains the HPD of the predictive density then the HPD ratio will be 1, suggesting that the two distributions are statistically consistent. Conversely, if the ratio is close to 0, it indicates little or no overlap, suggesting significant differences between the range of values for the state that would be inferred from these two distributions.

Figure 2(b) illustrates the computation of the HPD overlap. The HPD for the predictive distribution (top) and the normalized likelihood (middle) are computed, and the size of their intersection is normalized by the size of the smaller of the two HPDs (bottom).

2.2.3 Predictive Checks

The previous metrics compared two distributions of the state: one based on the accumulated past information and the other based on the isolated information from the likelihood of the most recent spiking observation. Another approach is predictive checking, which compares the observation itself to a predictive distribution of that observation estimated from the model. García and Castellanos [2007] proposed Bayesian checking of the second levels of hierarchical models with partial posterior predictive checking. Moran et al. [2019] used population predictive checking to evaluate model's goodness-of-fit. Guttman [1967] used future observations to evaluate model's goodness-of-fit. Predictive checks have the advantage that they allow for the computation of p-values for individual observations, which are easy to interpret [Meng, 1994, Gelman et al., 1996].

Predictive checks for spiking data can be challenging to implement, since in sufficiently small intervals, any spikes are unlikely compared to no spiking at all. One approach to deal with this might be to consider patterns of spikes over longer fixed intervals over which multiple spikes would be expected. Here, we take a simpler approach of focusing only on spike times and measuring the likelihood that an observed spike comes from a particular neuron (for spike-sorted models) or has a particular set of waveform features (for clusterless models).

In particular, let $y_{k,j}$ be the identity of the neuron (spike-sorted model) or a spike waveform feature (clusterless model) for the jth spike occurring in the kth time bin. We define the predictive distribution of the observation based on the state model as

$$f_{\text{pred}}(y_{k,j}) = \int p(y_{k,j} \mid x_k)\, p(x_k \mid y_{1:k-1})\, dx_k, \tag{3}$$

where $p(y_{k,j} \mid x_k)$ is the observation process and $p(x_k \mid y_{1:k-1})$ is the one-step prediction distribution of the current state given all previous spikes. We can sample new observations $\widetilde{y}_{k,j}$ from this predictive distribution, and compute the probability that the observed spike is at least as likely as the sampled spike:

$$p_{k,j} = \Pr_{\widetilde{y}_{k,j} \sim f_{\text{pred}}} \left(\log f_{\text{pred}}(y_{k,j}) \geq \log f_{\text{pred}}(\widetilde{y}_{k,j}) \right). \tag{4}$$

The log terms in Equation (4) are unnecessary, but suggest the method to estimate $p_{k,j}$. We sample x_k multiple times from the one-step prediction distribution, $p(x_k \mid y_{1:k-1})$, and use these to generate samples of the spike observations, $\widetilde{y}_{k,j}$, from the predictive distribution. We then compute the fraction of these samples whose log likelihood is smaller than or equal to that of the observed spike.

Figure 2(c) illustrates the computation of a predictive p-value for spike-sorted data. Many values of the state are sampled from the predictive distribution (top), each of which is used to sample a neuron that could have generated the observed spike (middle). The log likelihood of the observed spike is then compared to an empirical distribution of the log likelihoods of the sampled spikes (bottom) leading to a goodness-of-fit metric that can be interpreted as a p-value for that individual spike. High p-values suggest that the observed spike is consistent with the predictive distribution of spikes and a p-value near zero suggests that the observed spike is unexpected based on the predictive distribution computed from the past spiking and the model.

3 Simulation Study

Validating local goodness-of-fit metrics on real neural data is fundamentally limited: when the truth is unobserved, a metric that flags a moment of model failure cannot be distinguished from one that flags a moment of unusual neural activity. We therefore evaluate the three diagnostics—KL divergence, HPD overlap, and the rank-based predictive p-value—on a controlled benchmark in which the ground-truth fault at each moment is known by construction. The benchmark simulates a hippocampal place-cell population on a one-dimensional track and runs a Bayesian state-space decoder over the full session, deliberately perturbing either the generative process or the decoder's assumptions during four distinct windows. These four perturbations were chosen to highlight the distinction between model misfit that leads to decoding errors versus model misfit to which the decoding estimates remain robust. This relates to our original goal of identifying local periods where the information from the current spiking is inconsistent with the aggregated information from our model based on past spiking.

3.1 Generative Model

The simulation runs a Bayesian filter over a 32-second session whose generative process and decoder coincide everywhere except inside four misfit windows. Each window perturbs one specific component of the matched model.

Track and observation model. The state space is a one-dimensional linear track of length $L = 100$ arbitrary units (a.u.) discretized at unit spacing, $x \in \{0, 1, \ldots, 100\}$. The simulated population consists of eleven place cells with Gaussian place fields centered at $\mu_c \in \{0, 10, \ldots, 100\}$ and shared standard deviation $\sigma_{\mathrm{pf}} = 10$ a.u. The expected spike rate of cell c at position x is

$$\lambda_c(x) = \alpha\, \phi\Big(x \mid \mu_c, \sigma_{\mathrm{pf}}^2\Big), \tag{5}$$

where $\phi(\cdot \mid \mu, \sigma^2)$ is the Gaussian density and $\alpha = 5$ is a peak-rate scale factor. With a 1 ms simulation step, the peak rate is $\alpha/(\sigma_{\mathrm{pf}}\sqrt{2\pi}) \approx 200\,\mathrm{Hz}$. Spike counts are drawn independently across cells at each step, $y_{c,t} \sim \mathrm{Poisson}(\lambda_c(x_t))$.

Position dynamics. In baseline windows the animal's position evolves as a Gaussian random walk with step standard deviation $\sigma_{\mathrm{pred}} = 0.5$ a.u. The trajectory is reflected at 0 and 100 a.u. throughout the session, starts at $x_0 = 0$, and is continuous across phases: the final position of each phase becomes the initial position of the next.

Phase schedule. The 32-second simulation ($T = 32{,}000$ steps) is partitioned into eight contiguous phases: an opening baseline, four misfit windows, and three clean-recovery interludes that restore the matched-model configuration. Each misfit window perturbs either the generative process or the decoder:

Remap (steps 6,000**–**10,000**; decoder side):** the decoder evaluates per-spike likelihoods against a remapped place-field map in which three cell pairs ($0 \leftrightarrow 9$, $1 \leftrightarrow 8$, $2 \leftrightarrow 7$) have swapped centers, so the decoder's intended likelihood is at the wrong location for six of the eleven cells. Spike generation is unchanged.

History-dependent firing (steps 14,000**–**18,000**; generative side):** spikes are generated with a 1 ms hard refractory period followed by a 2–10 ms post-spike burst window in which the rate is multiplied by 3, consistent with the rough phenomenology of CA1 pyramidal-cell refractory and burst windows; the decoder continues to assume independent Poisson firing. The per-spike spatial likelihood for any individual spike is unchanged, so the misfit lives entirely in the spike-train temporal joint distribution.

Drift (steps 22,000**–**26,000**; generative side):** the trajectory is generated with persistent velocity, $v_t = 0.8\, v_{t-1} + \eta_t$ and $x_t = x_{t-1} + v_t$ with $\eta_t \sim \mathcal{N}(0, \sigma^2_{\text{pred}})$; the decoder continues to assume a memoryless random walk with step standard deviation σ_{pred}.

Wide-dynamics noise (steps 30,000**–**32,000**; decoder side):** the decoder is given an inflated transition matrix with step standard deviation $\sigma_{\text{inflated}} = 20$ a.u. (forty times the baseline), so its one-step predictive distribution spreads broadly across the track; spike generation is unchanged. This phase is engineered to inflate the KL divergence while leaving HPD overlap and the rank-based predictive p-value near baseline.

The three intermediate clean-recovery windows (steps 10,000–14,000; 18,000–22,000; 26,000–30,000) restore both the generative process and the decoder to the baseline configuration, providing within-session reference periods against which each metric's behavior during a misfit can be calibrated. With this schedule fixed, each diagnostic can be evaluated directly against a known ground-truth fault, and the three responses can be compared across faults that differ in which side of the model is perturbed.

3.2 Simulation Results

Figure 3a shows a realization of the simulation, the decoding results, and each of the three local goodness-of-fit metrics. The top panel shows the true position of the simulated rat in magenta and the predictive decoding distribution in black. The regions with white backgrounds are the periods when the model is correctly specified and each colored background indicates a period of model misspecification. The second panel shows the normalized likelihood of the data as a function of position at the spike times. The third panel shows the spiking from each of the eleven simulated neurons. Since these are sorted by their place field centers, the pattern of spiking aligns with the likelihood as a function of position. Note that the remapped period is the only one where the model misfit leads to incorrect decoding estimates; the other misspecified periods influence the decoding estimates in ways that increase the uncertainty of the estimates but do not lead to inconsistency between the likelihoods and the predictive distributions from the model.

The lower three panels of Figure 3a show, from top to bottom, the values of the HPD overlap, KL divergence, and the rank-based predictive p-value (as $-\log(p)$) for each spike. In each panel a horizontal line marks the threshold used to flag poor fit. For the HPD overlap, where low values indicate misfit, a spike is flagged when its value falls below the threshold, taken as the 1st percentile of the HPD overlap over the baseline periods pooled across 100 independent realizations of the simulation. For the KL divergence, where high values indicate misfit, a spike is flagged when its value exceeds the threshold, taken as the 99th percentile over the same pooled baseline. Pooling many realizations stabilizes these baseline-derived thresholds, which are otherwise noisy when estimated from a single run. For the predictive p-value, a spike is flagged when $p \leq 0.05$ (equivalently $-\log(p) \gtrsim 3$), a fixed cutoff rather than a baseline-derived bound. We see that all three metrics agree that there is lack of consistency during the remapping period, where the likelihood from the remapped cells is in a different part of the environment than the model estimate based on the incorrect place field estimates. For the period where history dependence is misspecified, the estimated decoding distribution becomes narrower, but remains consistent with the spike likelihoods, and none of the three metrics flag this period as leading to incorrect estimates. For the period where

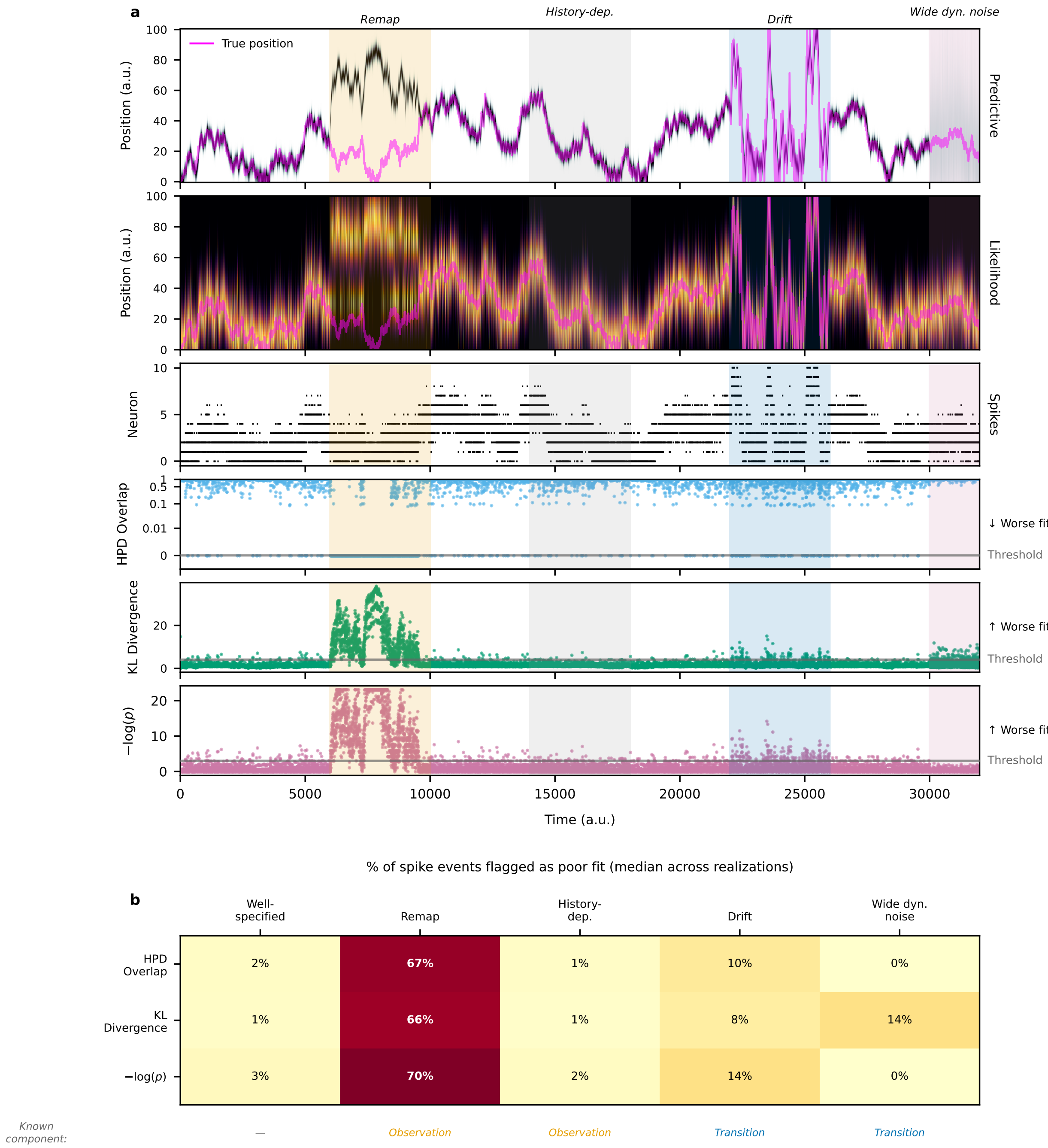

Figure 3: Per-spike goodness-of-fit diagnostics across a four-misfit simulation. The simulation walks through four model misspecifications separated by clean-recovery windows: place-field remapping, unmodeled history-dependent firing, persistent-velocity drift, and wide-dynamics noise. (a) The top block shows the predictive distribution as a heatmap of time and position. The position is overlaid as a magenta line. The misfit time periods are shaded in color. The second from the top block shows the per-spike likelihood as a heatmap, restricted to time bins containing at least one spike. The position is also overlaid as a magenta line. The third block shows the spike raster with neurons sorted by place-field peak position. The bottom three blocks show the three goodness-of-fit diagnostics (HPD overlap, KL divergence, and the rank-based predictive p-value) for each spike event, with a horizontal line marking the threshold used to flag poor model fit; the HPD overlap is plotted on a symmetric-log y-axis to resolve values near zero. (b) The percent of spike events flagged as poor fit by each diagnostic in each phase, reported as the median across 100 independent realizations of the simulation, with the well-specified "clean" recovery windows pooled into a single column for comparison. Each realization is scored against shared flag thresholds derived by pooling the baseline periods of all 100 realizations; pooling stabilizes both the thresholds and the per-phase fractions, which are noisy when estimated from any single run. A row beneath the heatmap labels each phase by the model component its misfit perturbs—the observation model (*Observation*) or the state-transition model (*Transition*)—with no induced fault (—) in the well-specified phases.

there is a drift term that is missed by the decoder model, the actual movement trajectory has notably increased velocity. The decoder only partially keeps up, so its predictive estimates lag behind the true location; this lag biases the predictive spiking distribution toward neurons consistent with the lagged position. As a result, the diagnostics flag a modest fraction of spikes during the drift period—most clearly the rank-based predictive p-value—though far less than during the remapping period. For the period where the decoder uses an inflated transition matrix, the predictive density becomes nearly uniform, losing all information from past spiking. Both the HPD overlap and rank-based predictive p-value measures indicate that this broad predictive distribution is consistent with the likelihood of each spike. However, this model misspecification highlights one of the main differences between these two metrics and the KL divergence. When the predictive distribution includes regions that are unlikely to correspond to the observed spike, the KL divergence is substantially increased. This demonstrates that the KL divergence does not quite meet the definition of consistency between distributions that we described above.

Figure 3b shows a heatmap of the fraction of spikes flagged by each metric in each period, summarized as the median across 100 independent realizations of the simulation. A spike is flagged whenever its diagnostic crosses the corresponding threshold in the direction of worse fit: below the 1st-percentile baseline value for the HPD overlap, above the 99th-percentile baseline value for the KL divergence, or at or below the fixed 0.05 cutoff for the predictive p-value. The median fraction flagged during the remapping period is large for all three metrics (66–70%), demonstrating that all three methods are able to identify inconsistent spiking information leading to incorrect state estimates; this fraction nonetheless varies widely from realization to realization because it depends on how often the simulated trajectory enters the remapped fields. All three metrics agree in not flagging many spikes during the history-dependence period; the drift period is flagged a bit more—most notably by the rank-based predictive p-value—but still far less than the remapping period. For the period where the decoding model incorrectly specifies the state transition, and therefore overly discounts previous spike information, the HPD overlap and rank-based predictive p-value metrics do not identify inconsistency between the spike likelihoods and the flat predictive distribution, while the KL divergence flags a median of 14% of the spikes. On one hand, this suggests that the KL divergence is sensitive to this true model misspecification; on the other hand, this suggests that the KL metric will flag spikes whenever the predictive distribution is uninformative, even when that distribution is consistent with the information from the current spike.

4 Real Hippocampal Data Analysis

4.1 Data collection and behavioral task

We examined the ability of these local goodness-of-fit metrics to identify periods of inaccurate decoding from place cells in rat hippocampus as it performed a spatial bandit task on a maze with three radially arranged Y-shaped foraging patches emanating from a center point. A schematic of the maze and the rat's movement in this environment are shown in Figure 4c. The details of this task, and data collection methods are described in detail in Comrie et al. [2024]. Briefly, the rat was trained to run between six reward ports located in three Y-shaped "patches", each with a different probability of reward. The probability of reward periodically switched after a time or trial limit. Twenty-three tetrodes were implanted in the dorsal hippocampus CA1 region. Spikes were detected by a 100 µV threshold on the recorded neural signal filtered 600 Hz to 6,000 Hz. [Add spike sorting information.]

4.2 Data analysis results

Figure 4 shows a comparison of decoding from this hippocampal spiking data under two state models. The first assumes that the state process moves continuously through space at all times, as would be true if the state only represented the rat's actual location. However, it is known that during periods where the rat is immobile, hippocampal place fields reactivate in sequences that replay past movement trajectories generating a neural representation that can jump to far from the rat's current location [Wilson and McNaughton, 1994, Karlsson and Frank, 2009]. We expect the continuous

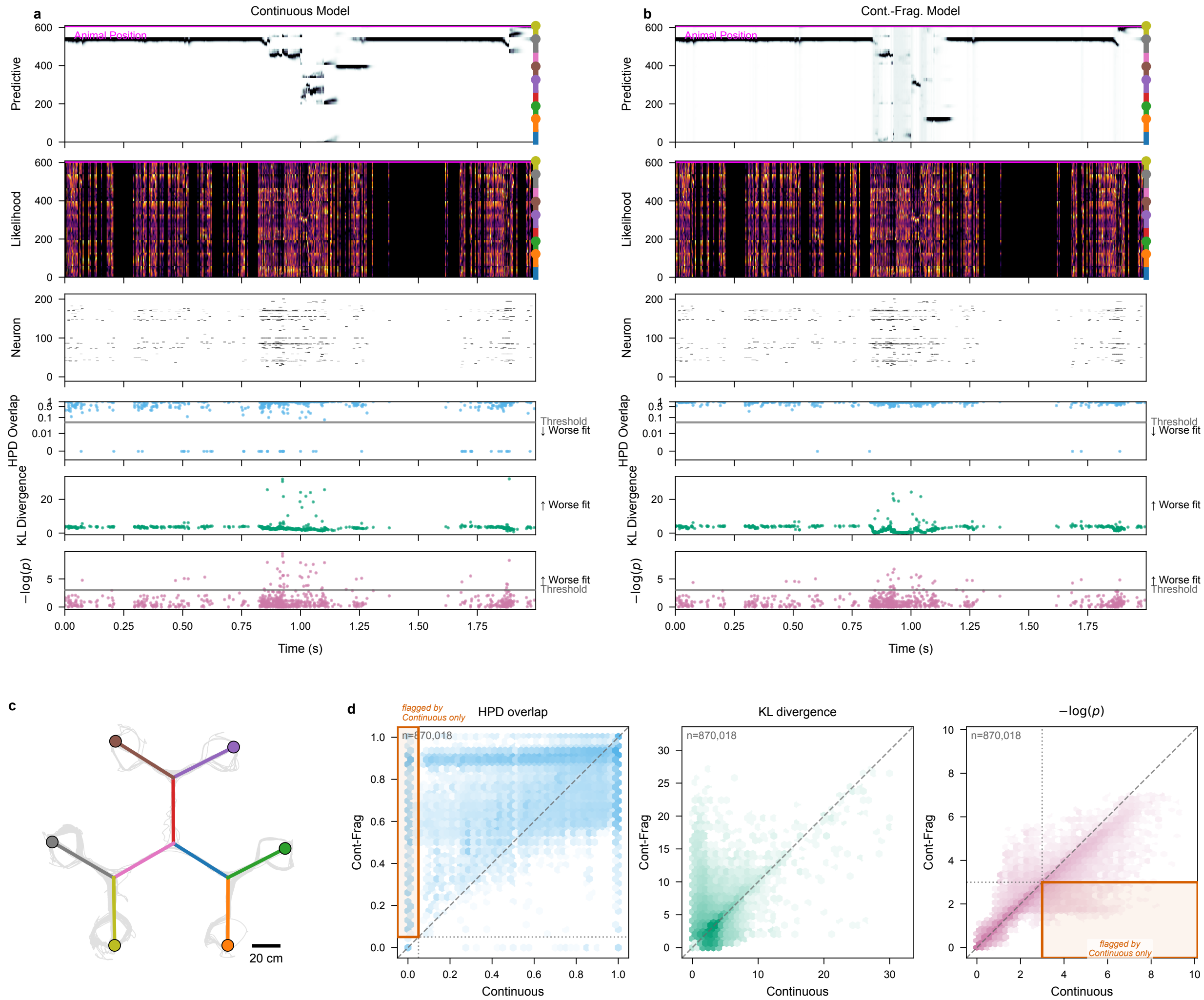

Figure 4: Per-spike goodness-of-fit diagnostics applied to a real hippocampal recording (203 simultaneously recorded cells) during navigation on a multi-arm maze; data from Comrie et al. [2024]. Two state-space decoders—a purely Continuous decoder and a Continuous–Fragmented decoder—are evaluated on the same session. **(a), (b)** A detail window containing a replay event, decoded with the Continuous and Continuous–Fragmented decoders, respectively. Within each of (a) and (b), all rows share a common time axis (x, seconds, relative to the start of the window). **Row 1, Predictive**: heatmap of the predictive posterior with linearized position on y and the animal's true linearized position overlaid. **Row 2, Likelihood**: same axes, restricted to time bins containing at least one spike, showing the combined per-spike likelihood. **Row 3, Neuron**: spike raster with neurons on y, sorted by place-field peak position. **Rows 4–6 (HPD overlap, KL divergence, $-\log(p)$)**: one marker per spike event placed at its spike time, with the per-spike diagnostic value on y; for the HPD overlap and the predictive p-value a horizontal line marks the threshold used to flag poor model fit (a fixed 0.05 cutoff), the KL divergence has no natural fixed cutoff and is shown without a threshold, and the direction of worse fit is labeled at the right. **(c)** 2D layout of the track, with the maze arms drawn as colored segments, colored dots marking the six reward wells at the arm ends, the animal's trajectory overlaid in grey, and a 20 cm scale bar. **(d)** Pairwise comparison of per-spike diagnostic values between the two decoders across the whole session, one hexbin panel per metric (HPD overlap, KL divergence, $-\log(p)$); x-axis is the Continuous decoder, y-axis is the Continuous–Fragmented decoder, and the dashed line is identity. For the HPD overlap and $-\log(p)$ panels, dotted lines mark the per-metric flag threshold on each axis (the same 0.05 cutoff used in panels (a)–(b)) and the outlined box marks the quadrant of spikes flagged by the Continuous decoder but not the Continuous–Fragmented decoder; the KL divergence panel has no threshold. Hexagons are colored by spike-event density (log scale).

state model to fail precisely at those times when the neural representation jumps. We compare this with an alternate state model called a continuous-fragmented model that allows the hippocampal representation to jump at discrete times. We have previously used this model structure to capture replay and other nonlocal hippocampal representations [Denovellis et al., 2021].

Figure 4a shows a brief segment of the decode using the continuous state model during a period when the rat is immobile. The panels from top to bottom are the same as in Figure 3a. While the rat rests (magenta line), we find a classic replay event with rapid transitions between areas of the maze. During this replay event, the continuous state model is unable to keep up and the predictive distribution consistently lags behind the information coming from the likelihood of the observed spikes. In the lower three panels, we see many of the spikes flagged as inconsistent with the predictive model estimates by all three goodness-of-fit metrics.

The contrast between decoding with the continuous and continuous-fragmented state models is shown in Figures 4a and b. We see that the likelihood for each spike is identical for these models (second panel from top) but the continuous-fragmented model decodes a rapidly moving state trajectory that the continuous model is unable to track. This happens because the predictive distribution becomes uniform or nearly uniform at time steps where the representation is estimated to be fragmented and the likelihood of subsequent spikes can then cause a jump in the state estimate. For the HPD overlap metric (fourth panel from top, blue), we see that nearly all of the spikes flagged in the continuous state model are no longer flagged when using the continuous-fragmented model that can track rapid jumps in the state process. For the rank-based predictive p-value (bottom panel, pink), we see that some of the $-\log(p)$ values are reduced for the continuous-fragmented relative to the continuous state model. This is because the continuous state model leads to predictive densities that are concentrated in the wrong location, which makes the observed spiking very unlikely, while the continuous-fragmented state model leads to broad predictive distributions during replay events, which makes the observed spikes still unlikely but less so. For the KL divergence metric (fifth panel from top, green), we find that many spikes retain high values for the continuous-fragmented model. This is consistent with our simulation results, where nearly flat predictive distributions led to increased KL values, even though they are consistent with the spiking likelihood according to our earlier definition.

Figure 4d shows hexbin density plots comparing the values of each of the local goodness-of-fit metrics for the continuous state model versus the continuous-fragmented state model applied to each spike, with each hexagon colored by the number of spike events it contains. The leftmost panel shows the values of the HPD overlap metric in blue. Here, hexagons above the 45° line correspond to spikes for which the continuous-fragmented state model produces predictive distributions more consistent with the spiking data than the continuous state model. Of note here is the dense vertical band at $x = 0$ and $y > 0$, corresponding to spikes for which the predictive distribution from the continuous state model has no overlap with the spike likelihood but for which the predictive distribution from the continuous-fragmented state model has nonzero overlap. There is no corresponding density at $y = 0$ and $x > 0$, suggesting that the HPD overlap is successful at identifying times when the continuous state model is inconsistent with the observed spiking and that adding the fragmented state process into the model alleviates this inconsistency. Quantitatively, of the 18,328 spikes that the continuous model flags under the HPD overlap, 17,123 (93%) are no longer flagged by the continuous-fragmented model (which we call *rescued*), while only 139 spikes are flagged by the continuous-fragmented model but not the continuous one (*newly flagged*).

The center panel of Figure 4d shows values of the KL divergence in green. In this case, hexagons below the 45° line correspond to spikes for which the continuous-fragmented state model produces predictive distributions more consistent with the spiking data than the continuous state model. There is indeed a large concentration of density below this line in the region where the value for the continuous model is below 5. More striking though is the region above the 45° line where the value for the continuous model is around 2 and the value for the continuous-fragmented model goes up to 20 or so. As seen in the simulation, this corresponds to situations where the predictive distribution from the continuous-fragmented model is broad, as occurs whenever a fragmented state is estimated. This suggests that the KL divergence is less useful in identifying inconsistency from the continuous-fragmented model.

The right panel of Figure 4d shows the value of the negative log of the rank-based predictive p-value ($-\log(p)$) in pink. Once again, hexagons below the 45° line correspond to spikes for which the continuous-fragmented state model produces predictive distributions more consistent with the spiking data than the continuous state model. Of note here is the band of density below this line that extends beyond a value of 3 on the x-axis (corresponding to a p-value below 0.05, since $-\log(0.05) \approx 3$). There is no corresponding density above the line with a value for the continuous-fragmented model above 3. As with the HPD overlap metric, this suggests that the rank-based predictive p-value is also successful at identifying times when the continuous state model is inconsistent with the observed spiking. Quantitatively, the continuous-fragmented model rescues 8,786 of the spikes flagged by the continuous model under the predictive p-value while newly flagging only 1,654.

5 Discussion

State space modeling has become a powerful tool for describing the latent dynamics of neural systems. However, validating these models is inherently difficult because latent states are unobservable and many models are purposefully simplified. To address this, we developed multiple local goodness-of-fit metrics that compare the instantaneous information received from each spike to the aggregated information computed based on recent spiking activity and the state-space model assumptions. Rather than dismissing a model in its entirety, these metrics are able to assess the consistency between immediate observations and the information accumulated through the state-space structure, providing a more granular assessment of decoding reliability.

We defined three classes of metrics: F-divergence measures (e.g., KL divergence), overlap ratios between credible regions (HPD overlap), and predictive checks. Although KL divergence is computationally simple, it is sensitive to distribution tails and lacks a clear upper bound to assess similarity. In contrast, the HPD overlap offers a more robust geometric measure of similarity, consistently maintaining lower error rates across various conditions. Predictive checks provide a direct statistical assessment through observation-level p-values, but are more computationally intensive. Although each metric has particular advantages, when used in combination, they provide a more complete perspective on decoding reliability.

We illustrated the application of these metrics to simulated and real spike-sorted data from rat hippocampus. These same metrics naturally extend to other neural data types when incorporated into state space models. In particular, recent work has shown that clusterless neural spiking models can substantially reduce bias and variance in neural decoding and estimation problems when compared to spike-sorted models [Deng et al., 2016]. The metrics, as defined in Section 2, naturally extend to clusterless models. In that case, a lack of fit for any particular spike suggests that a spike with that waveform was unexpected based on the accumulated information from the state space model up to that point in time.

For simplicity, in this paper, we focused on comparing instantaneous information in the likelihood of each spike with aggregated information from the past contained in the one-step prediction density. A natural and powerful extension of these methods is obtained by replacing the one-step prediction density with any posterior density derived from the state-space framework. In particular, replacing the one-step prediction density with a smoothing density (e.g., from a fixed-interval smoother over the observation period) would allow us to leverage both past and future data to increase statistical power in identifying model misspecification.

One potential challenge for applying these methods broadly has to do with the computational burden of computing these metrics as the dimensionality of the state process increases. In all of our examples, we computed the KL divergence measure and the HPD overlap by partitioning the state space into a lattice and computing the likelihoods and prediction densities at each point in that lattice. As the dimension of the state increases, the size of that lattice will grow exponentially. While the predictive checks are the most computationally burdensome in lower state dimensions, the fact that we are sampling from the state-space rather than computing values over a lattice means that in higher dimensions the predictive checks will become more computationally efficient. It also suggests that we may be able to improve the efficiency of the KL and HPD overlap metrics by using sampling-based estimation methods. This will not solve the curse of dimensionality, as the number of samples required for accurate estimates will likely still grow with dimension, but it is likely to allow for efficient estimation in moderate dimensions. Another approach for dealing with this issue may be to compute these consistency metrics in lower-dimensional subspaces of the state space. Identifying subspaces that retain the statistical power of the original space is a topic for future exploration.

Another potential challenge relates to identifying time-scales for which to assess for lack of fit. These methods allow us to focus on time-scales as short as individual spikes, and for our place cell decoding examples individual spikes are sufficiently informative that we are often able to detect local model misfit at this time-scale. For other neural systems, it may be the case that local model errors are more likely to be detected over longer time-scales. In this paper, we discuss how to extend these goodness-of-fit measures to longer time intervals, but determining what time-scales are most appropriate may require relying on prior knowledge about the system, or exploring the values of these metrics across multiple time-scales.

We believe that these local goodness-of-fit tools provide an important step toward making inferences from state-space neural models more reliable and interpretable. By pinpointing specific moments when the model assumptions break down, researchers can be more confident in their inferences. Ultimately, these measures empower us to refine our understanding of neural dynamics and deepen our understanding of how the brain encodes and processes information.

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